

sources operate at different voltage and impedance levels. While conventional microwave sources typically operate on relatively low-voltage and high-impedance levels, HPM sources operate at relatively higher-voltage and lower-impedance levels.

HPM sources. HPM sources are broadly classified as impulsive sources and linear beam sources. In the case of impulsive sources, pulsed microwave energy is generated by charging the antenna, a transmission line or a tuned circuit directly, and making these ring for one or several cycles by closing a switch.

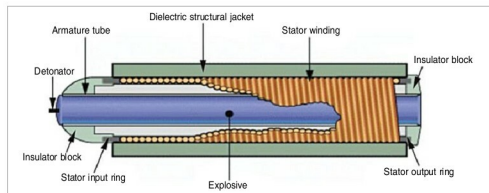


Fig. 4: Flux-compression generator

In the case of linear beam sources, microwave energy is generated by converting kinetic energy of an electron beam into electromagnetic energy of the microwave beam. While various ultra-wide-band sources are examples of impulsive sources, examples of linear beam sources include klystrons, travelling-wave tubes, backward-wave oscillators, magnetrons, cross-field amplifiers, split-cavity oscillators, virtual-cathode oscillators, gyrotrons, free-electron lasers and orbitron microwave masers.

Antennae. An antenna acts as an interface between the transmitter output and the medium which the radiated electromagnetic waves have to propagate through. In the case of an HPM weapon system too, it is an interface between the HPM source output and the surrounding atmosphere. Antennae play a crucial role

in the HPM system design. Some of the important factors that need to be addressed by HPM antennae include directivity, ultra-wide bandwidth, feed-to-antenna coupling efficiency and compactness.

Directivity plays an important role. Since the HPM equipment is in close proximity to the antenna, the equipment may suffer damage if radiated energy is not radiated effectively. An HPM signal, due to its very short pulse duration, inherently has very wide bandwidth. The antenna therefore has to meet the stringent requirements of handling this ultra-wide bandwidth.

Another important requirement is that of proper matching between the feed and the radiating element, lest the resultant standing waves would cause voltage breakdown. HPM antennae tend to be massive in order to avoid voltage breakdown at operating electric field levels. To meet the increasing requirements of having HPM weapon systems on smaller platforms, antenna size can play an important role.

Antenna shape is also an issue as it influences to a great extent whether air breakdown phenomenon is an issue or not at high power levels. Horn antennae and antenna arrays are the promising candidates for HPM sources with the former presently being the most commonly-used type. Different types of horn antennae including conical, circular, rectangular, corrugated and half-oval, and TEM horn antennae are used

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